

## ECE-UY 3613, Spring 2024, Quiz 5

Name:

ID:

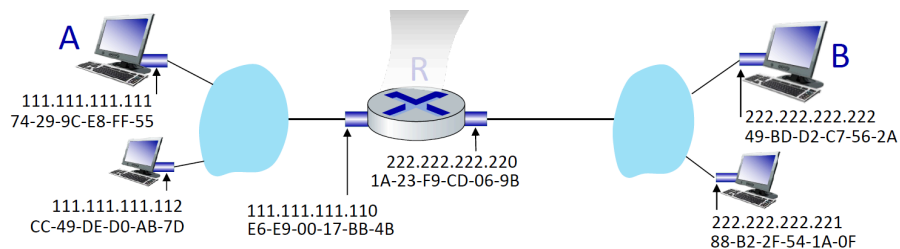
1. Given a data bit sequence of 10101100111 and generator sequence of 1101, what is the CRC code? What is the transmitted sequence with CRC?

CRC = 110

transmitted sequence: 1010110011110

2. For the network in the following figure,

- when A sends out an IP packet to 111.111.111.112, what is the source and destination MAC addresses of the data frame sent out by A?
- when A sends an IP packet to B, what is the destination MAC address in the data frame sent out by A, what is the source MAC address in the data frame received by B?



a. ① 74-29-9C-E8-FF-55  
② CC-49-DE-D0-AB-7D

b. ① E6-E9-00-17-BB-4B  
② 1A-23-F9-CD-06-9B

①

$$\begin{array}{r} \begin{array}{ccccccccc} & & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ \hline 110 & \overline{) 100110011000} & & & & & & & & & & \\ & 1101 & & & & & & & & & & \\ \hline & & 1111 & & & & & & & & & \\ & & 1101 & & & & & & & & & \\ \hline & & & 1010 & & & & & & & & \\ & & & 1101 & & & & & & & & \\ \hline & & & & 1110 & & & & & & & \\ & & & & 1101 & & & & & & & \\ \hline & & & & & 1111 & & & & & & \\ & & & & & 1101 & & & & & & \\ \hline & & & & & & 1010 & & & & & \\ & & & & & & 1101 & & & & & \\ \hline & & & & & & & 1110 & & & & \\ & & & & & & & 1101 & & & & \\ \hline & & & & & & & & 110 & & & \end{array} \end{array}$$